1.What are the Criteria to be a Genetic Code?

Characteristics of the Genetic Code

The genetic code itself, which translates information from nucleotide sequences into proteins, is defined by several specific properties:

- **Triplet Nature:** The code uses four nucleotide bases (A, C, G, and T) arranged in **three-letter "codons"** to specify amino acids.
- **Universality:** The code is generally **universal**, meaning the same codons specify the same amino acids across different living organisms.
- **Degeneracy:** The code is degenerate, meaning **multiple codons** can code for the same amino acid. Out of 64 possible codons, 61 code specifically for amino acids.
- **Non-overlapping and Comma-less:** The genetic code is read as a continuous sequence **without overlapping** or gaps between codons.
- **Non-ambiguous:** Each specific codon codes for **only one** particular amino acid.

2.Why do we differ from our parents? Remarks on the origins of genetic diversity?

We differ from our parents primarily due to the processes of **meiosis and fertilization**, which introduce significant genetic variation into offspring. The origins of this genetic diversity can be attributed to several key biological mechanisms:

- **Crossing Over (Recombination):** During the first stage of meiosis, homologous chromosomes align closely in a process called synapsis. They exchange segments of DNA at points called chiasmata, resulting in **recombinant chromosomes** that contain a unique mixture of maternal and paternal genetic material.
- **Independent Assortment (Random Orientation):** During metaphase I of meiosis, homologous chromosome pairs orient themselves randomly at the cell's equator. Since each pair "flips a coin" to decide which chromosome goes to which daughter cell, a single human can produce over **8 million genetically distinct gametes**.
- **Random Fertilization:** The combination of a unique egg from the mother and a unique sperm from the father creates a zygote with a genotype that is statistically **one-in-a-trillion**, ensuring that no two siblings (except identical twins) are genetically the same.
- **Mutations:** Genetic diversity also arises from **mutations**, which are changes in the DNA sequence caused by errors during replication or exposure to environmental mutagens. These mutations can occur in somatic cells or be passed down if they occur in germ cells.
- **Genomic Alterations:** Larger-scale changes, such as **duplications, deletions, inversions, and translocations**, contribute to the creation of genetic novelty and genome evolution.
- **Horizontal Gene Transfer:** In some organisms, particularly microbes, genetic material can be transferred between different species, further increasing genetic variety.

Through these mechanisms, meiosis and fertilization reshuffle gene variants (alleles) into new combinations, providing the raw material for **natural selection** and allowing populations to adapt and survive over the long term.

3.What are the basic difference of the prophase of mitosis and prophase I of meiosis?

The basic differences between **prophase of mitosis** and **prophase I of meiosis** center on the complexity of the process and the introduction of genetic variation. While both involve the condensation of chromatin into visible chromosomes and the breakdown of the nuclear envelope, prophase I of meiosis includes unique sub-stages and events that do not occur in mitosis.

Feature	Prophase (Mitosis)	Prophase I (Meiosis)
Type of Division	Part of mitotic (equational) division.	Part of meiotic (reductional) division.
Synapsis	**Absent.** Homologous chromosomes do not pair up.	**Present.** Homologous chromosomes pair closely to form **bivalents or tetrads**.
Crossing Over	**Absent.** No exchange of genetic material occurs.	**Present.** Genetic recombination occurs between non-sister homologous chromatids.
Chiasmata	**Not formed.**	**Formed.** These are the visible points of contact where crossing over takes place.
Genetic Variation	No new variation is introduced; daughter cells remain identical.	**Generates genetic variation** by creating new combinations of maternal and paternal genes.
Chromosome Number	Maintains the **diploid (2n)** number.	Starts as diploid (2n) but **prepares for reduction** to haploid (n).
Purpose	To prepare the cell for an equal division of identical genetic material.	To prepare for reduction division and ensure genetic diversity in gametes.

4. What is an oncogene? How do they cause tumor or cancer?

An **oncogene** is a gene that has the potential to cause cancer. These genes are mutated or activated versions of **proto-oncogenes**, which are normal genes that help cells grow, divide, and stay alive.

How Oncogenes Cause Tumors and Cancer

- **Activation through Mutation:** A proto-oncogene becomes an oncogene when it **mutates** or when there are **too many copies** of it. This causes the gene to be "turned on" or activated when it is not supposed to be.
- **The "Gas Pedal" Analogy:** In normal cell function, a proto-oncogene works like a **car's gas pedal**, helping the cell grow and divide at a controlled pace. An oncogene is compared to a **gas pedal that is stuck down**, causing the cell to divide rapidly and without regulation.
- **Loss of Growth Control:** When these genes are activated, cells **lose the ability to control their growth**. This uncontrolled proliferation results in the formation of **tumors**, which are masses of cells that can damage or invade surrounding healthy tissue.

- **Interaction with Other Genes:** Cancer often involves a combination of "stuck" gas pedals (oncogenes) and **failed brakes** (mutated tumor suppressor genes like **TP53**). While oncogenes drive growth, the loss of tumor suppressor genes means the cell can no longer halt the cycle to repair DNA damage or undergo programmed cell death.

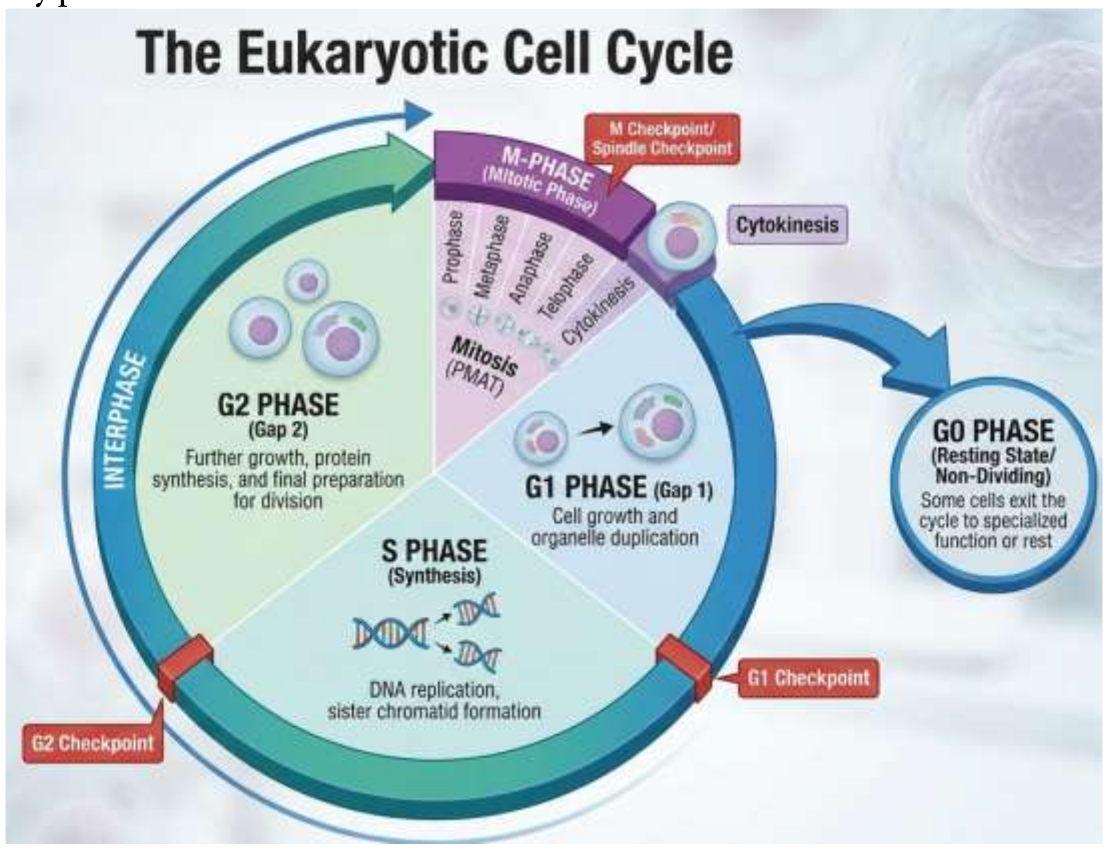
Common Examples of Oncogenes

According to the sources, some frequently identified oncogenes include:

KRAS C-Myc EGFR HER2 PIK3CA

5. Explain the different stages of the cell cycle with a diagram.

The cell cycle is a repeating series of events involving cell growth, DNA synthesis, and division. It is divided into two primary phases:



1. Interphase (Longest Phase, ~90%)

- **G1 Phase (First Gap):** The cell grows and performs normal biochemical functions while accumulating building blocks for DNA.
- **S Phase (Synthesis): DNA replication** occurs, resulting in two identical sister chromatids.
- **G2 Phase (Second Gap):** The cell replenishes energy, synthesizes proteins, and makes final preparations for division.

2. Mitotic Phase (M-Phase)

- **Mitosis:** Nuclear division consisting of four stages: **Prophase, Metaphase, Anaphase, and Telophase (PMAT).**
- **Cytokinesis:** The final physical separation of the cytoplasm into **two identical daughter cells**.

Additional States & Regulation

- **G0 Phase:** A resting state where cells (like nerve cells) exit the cycle and do not divide.
- **Checkpoints:** Internal regulators near the end of **G1, G2/M transition, and Metaphase** ensure the cell is ready to proceed without mutations.

6. How cell cycles are regulated and indicate the major checkpoints where regulation occurs.?

The regulation of the cell cycle is a precise process controlled by both **internal and external mechanisms** to ensure that daughter cells are exact duplicates of the parent cell. This regulation is vital because mistakes in chromosome duplication or distribution can lead to mutations or uncontrolled growth, such as cancer.

1. Types of Regulators

- **Internal Regulators:** These respond to events inside the cell. A key protein involved in this process is **Cyclin**, which regulates the cycle in eukaryotic cells. Internal controls ensure that a cell does not proceed to a new stage (like mitosis) until all previous requirements (like DNA replication) are successfully met.
- **External Regulators:** these respond to signals from outside the cell. They include **growth factors** that can speed up or slow down division, the death of a nearby cell, or physical crowding, which can inhibit further division.

2. Major Checkpoints

Checkpoints are specific points in the cycle where progression can be halted until conditions are favorable. The three major checkpoints are:

- **G1 Checkpoint (Restriction Point):** Located near the end of the G1 phase, this is the primary point where the cell commits to division. It is influenced by external signals like **growth factors and nutrients**, as well as internal factors like **cell size and DNA damage**.
- **G2 Checkpoint (G2/M Transition):** Occurring at the end of the G2 phase, this checkpoint ensures that all chromosomes have been **replicated** and that the replicated DNA is not damaged before the cell enters the mitotic phase.
- **M Checkpoint (Metaphase-Anaphase Transition):** This occurs during the mitotic phase (specifically metaphase). It prevents the cycle from proceeding to anaphase until all **chromosomes are properly attached to the spindle fibers**, ensuring equal distribution to daughter cells.

Failure in these regulatory mechanisms, such as damage to the **p53 gene**, can lead to cells losing their ability to control growth, resulting in the formation of **tumors**.